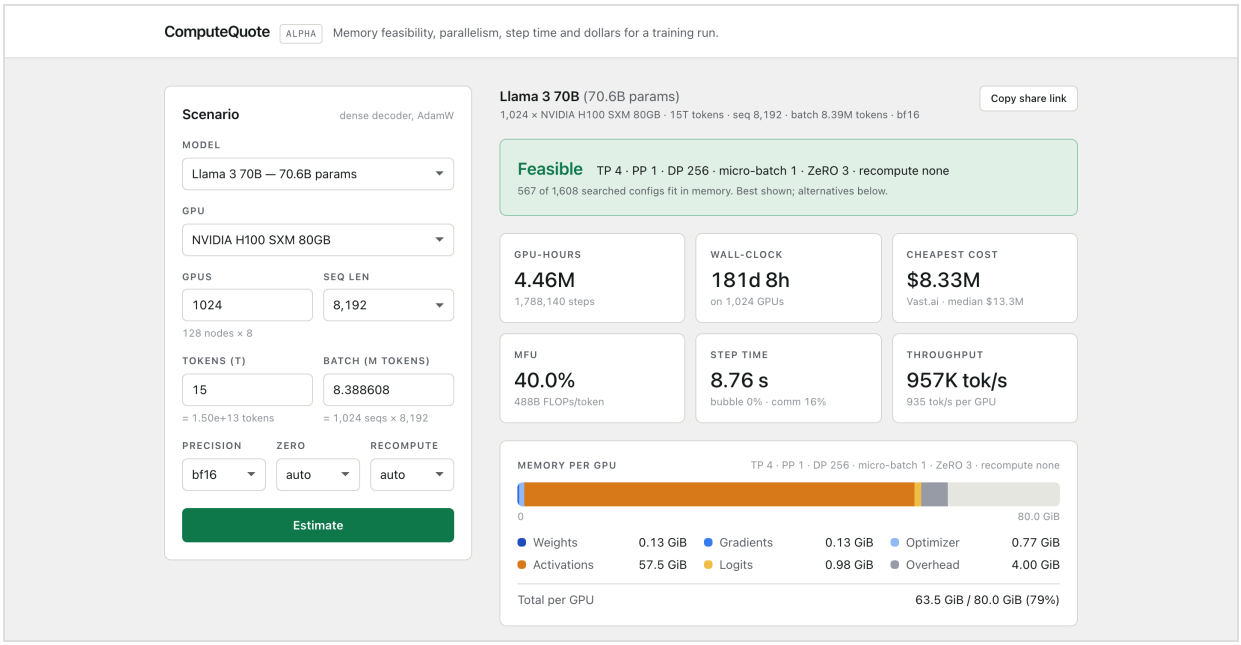
ComputeQuote — Alpha Release & Task Backlog

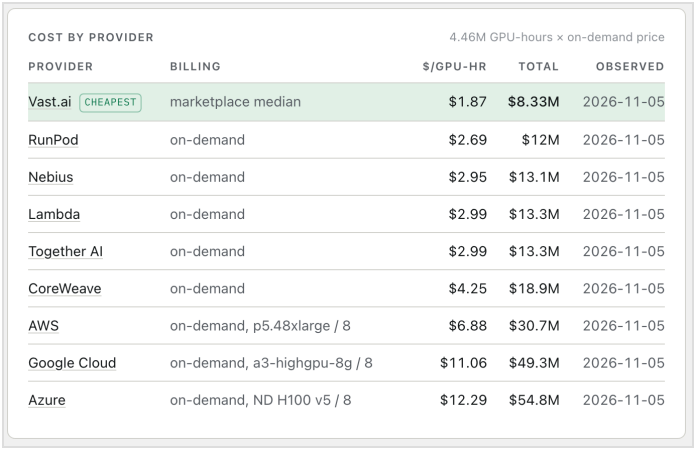
CS146S Aut26 · Yogesh Seenichamy · November 5, 2026 · Repo: github.com/YogeshSeeni/computequote · Backlog: github.com/users/YogeshSeeni/projects/2

1. Early working demo

ComputeQuote predicts what an AI training run will cost from first principles. The alpha takes a dense-transformer architecture (preset or custom), a GPU type and count, a token budget and a global batch, then: (1) counts parameters, (2) enumerates every tensor/pipeline/data-parallel split with micro-batch, ZeRO stage and recompute mode, (3) rejects configs that do not fit in GPU memory, with a reason, (4) ranks the rest with a roofline-plus-communication step-time model, and (5) converts the winner into GPU-hours, wall-clock and dollars across a price table. Every estimate is a shareable URL. The stack from the technical spec runs locally: a pure-Python engine (`engine/`, no I/O, 115 tests), a FastAPI service (`api/`) and a Next.js page (`web/`); `make api` and `make web` start it.



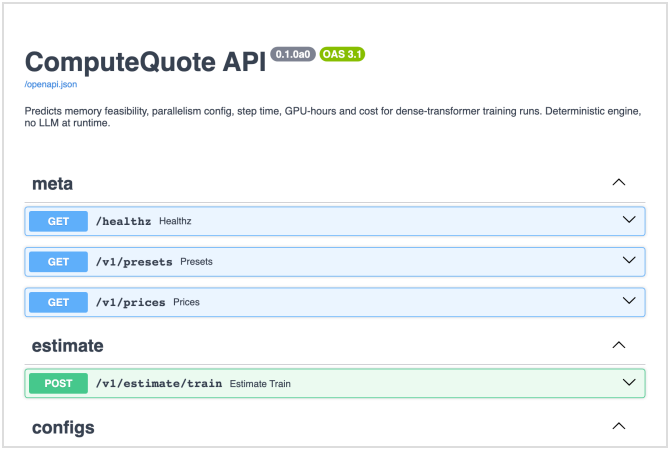
Calculator: Llama 3 70B, 15T tokens, 1,024 × H100. Best of 1,608 searched configs (TP 4, ZeRO-3, 40% MFU), 63.5 of 80 GiB per GPU, 4.46M GPU-hours, 181 days. The URL encodes the scenario.



COST BY PROVIDER — 4.46M GPU-hours × on-demand price

PROVIDER	BILLING	$/GPU-HR	TOTAL	OBSERVED
Vast.ai CHEAPEST	marketplace median	$1.87	$8.33M	2026-11-05
RunPod	on-demand	$2.69	$12M	2026-11-05
Nebius	on-demand	$2.95	$13.1M	2026-11-05
Lambda	on-demand	$2.99	$13.3M	2026-11-05
Together AI	on-demand	$2.99	$13.3M	2026-11-05
CoreWeave	on-demand	$4.25	$18.9M	2026-11-05
AWS	on-demand, p5.48xlarge / 8	$6.88	$30.7M	2026-11-05
Google Cloud	on-demand, a3-highgpu-8g / 8	$11.06	$49.3M	2026-11-05
Azure	on-demand, ND H100 v5 / 8	$12.29	$54.8M	2026-11-05

Same scenario priced per provider ($8.3M–$54.8M), cheapest highlighted, observation dates shown.



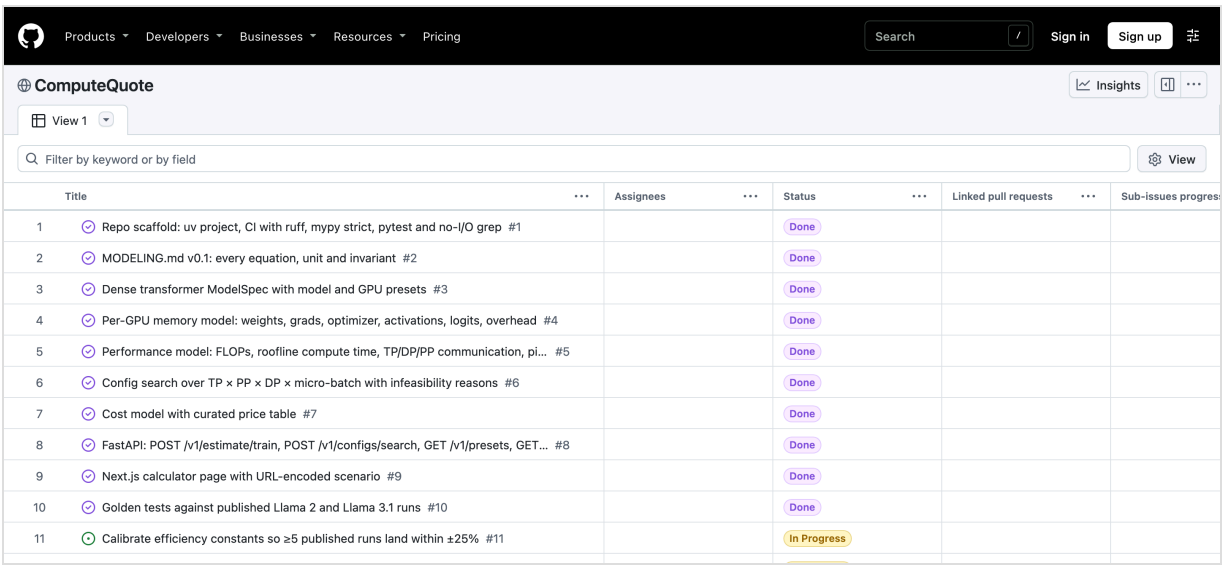
FastAPI service behind the page; the Pydantic schemas double as docs for teammates and agents.

Feature (proposal MVP list)	Status	Notes
Model spec input + presets	**Functional**	8 presets (Llama 2/3, Mistral, Qwen) within 1% of model-card parameter counts; custom specs via API only

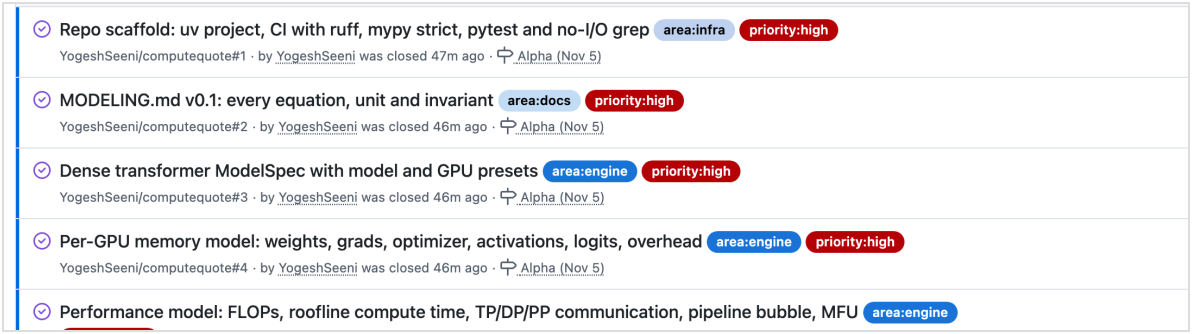
Feature (proposal MVP list)	Status	Notes
Memory model	**Functional**	Weights/grads/optimizer with ZeRO 0–3; activations for none/selective/full recompute under TP + sequence parallel; fp32 logits; fixed overhead
Config search	**Functional**	1,608 candidates for the 70B scenario in 23 ms; infeasible ones say “needs X GiB, GPU has Y GiB”
Performance model	**Functional**	6N + attention FLOPs, roofline vs peak and HBM, TP/DP/PP communication, 1F1B bubble, MFU. No interleaved pipeline yet
Cost + shareable URL	**Partial**	Exact-cent cost across 9 providers from a **curated, hand-entered price table** (approximate list prices, dated, staleness flags). Share links are query strings; short IDs are backlog
Validation vs published runs	**Partial**	6 runs in `engine/validation.py` (below); no public validation page yet
Deploy, live prices, Postgres, inference, MoE, share cards	**Placeholder**	Not started; in the backlog

Validation snapshot (engine v0.1, predicted ÷ reported GPU-hours): Llama 2 7B **0.92**, 13B **0.86**, 70B **0.96**; Llama 3.1 405B **1.11**; Llama 3.1 8B 0.38 and 70B 0.67 (misses). Four of six are inside the ±25% target. The misses are Llama 3.1 model-card figures that count every training stage, not only the 15T-token pretraining run the engine models; cluster sizes are assumed because they are unpublished. The one efficiency constant was calibrated once on this table, from 0.70 to 0.55.

2. Documented backlog



Public GitHub Project board (33 issues) with status per item.



The same issues with priority and area labels and milestones (Alpha Nov 5 / Final Dec 10).

Board: github.com/users/YogeshSeeni/projects/2. Each issue has a title, description, status (Todo / In Progress / Done), a `priority:high|medium|low` label, an area label and a milestone; the whole backlog is generated by `scripts/backlog.sh` so it is reproducible. Single-person team, so no assignee. Counts: **Done 10** (all high), **In Progress 2** (calibration; short links), **Todo 21** (8 high, 9 medium, 4 low). Highest-priority next steps: calibrate on more runs and model interleaved pipelines; deploy the API (Fly.io) and web app (Vercel); replace the curated price table with nightly Vast.ai and RunPod collectors into Postgres; ship the validation page.

3. Reflection on AI/LLM tool integration

Tools used. Claude Code (Fable 5.1) for everything so far: plan mode turned the three project PDFs into an approved build plan; three parallel subagents implemented the memory model, the performance model and the Next.js page concurrently; the main session did cost, search, the API, calibration and this document. The GitHub CLI created the repo, labels, milestones, issues and board from a script. No LLM runs inside the product; the engine is deterministic.

How it changed the process. The spec-first workflow from the proposal held: `docs/MODELING.md` was written before any engine code, every function cites its section, and subagents were briefed with exact signatures and a TDD requirement. Three modules landed in about 40 minutes with 73 hand-derived tests and no interface mismatch except one introduced deliberately mid-build (uneven pipeline stages), which both engine agents flagged in their reports. Empty folder to running stack with 115 tests took one evening.

Successes. The engine agents cross-checked arithmetic with standalone scripts and documented two spec ambiguities instead of guessing. Hypothesis property tests (memory non-increasing in TP/PP/DP/ZeRO) came almost free once invariants were in the spec. Calibration was a five-line sweep because the engine has no I/O.

Challenges. The model recalled GPU specs, provider prices and published GPU-hour figures from memory, so the price table is labelled approximate and the validation table carries assumed cluster sizes; both need human verification before any real decision, which is why live collectors and a public validation page are high priority. Unit discipline needed a rule (`memory_bytes`, `step_time_s`) plus a mypy strict gate. And the agents are optimistic by default: the first efficiency constant gave 45–55% MFU where published runs imply 25–43%, and only the golden tests caught it.

Insight. Treating the modeling spec as the contract between human and agents beat reviewing code line by line: when a number looked wrong, the fix was a spec sentence plus a test, and the agents did the rest.

4. Communication & presentation

How progress is communicated. Requirements and status reach humans and agents through the same artifacts: `docs/MODELING.md` (equations, units, invariants), `CLAUDE.md` (layout and rules), `docs/prompts.md` (prompt log with before/after results), the public project board, and the README's three-command run instructions. For non-technical readers the calculator is the status report: a green or red verdict, six headline numbers and a cost-by-provider table. Stakeholders get this document and share links, not code.

Demo-day talk (5–8 min, one live scenario), unchanged from the proposal and mapped to alpha status:

- **Context (1 min):** why "6 × params × tokens ÷ peak" spreadsheets go wrong, and for whom. *Ready.*
- **Demo (3 min):** estimate a 70B run end to end, show the memory verdict and config, open its share link, compare providers, show the validation page. *Calculator and share link ready locally; validation page and deployed URL are backlog (#13, #18).*
- **AI integration (1–2 min):** the spec-driven agent workflow with before/after prompts from `docs/prompts.md`. *Ready; the log grows each milestone.*
- **Reflection (1–2 min):** results vs success criteria (±25% on ≥5 runs, p95 < 1 s, ≥8 providers, < 2 min to an estimate), where AI helped or failed, roadmap. *Today: 4/6 runs within tolerance, 23 ms search, 9 curated providers; live prices and more runs needed by Dec 10.*

The final deliverable adds the deployed app so the audience can open the share link on their own devices during the talk.